Named as a Hazard, Left Unlimited

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CITE AS

Pendergrass K. "Named as a Hazard, Left Unlimited". Heavy Metal Certified, 2026. https://heavymetalcertified.com/briefings/the-prop--65-gap. CC BY 4.0.

CORRESPONDENCE

Corrections are logged publicly on the article page, each with its date, the affected value, and the source that resolved it.

KEYWORDS

heavy metals, food safety, dietary exposure

PUBLICATION HISTORY

This revision 2026-08.

Abstract

California is the one American jurisdiction that has written into law that these metals cause cancer or reproductive harm. Under Proposition 65 it lists lead, cadmium, inorganic arsenic, hexavalent chromium, and nickel — and then, as a right-to-know statute, sets no maximum level for any of them in food. For inorganic arsenic and nickel it adopts no oral safe-harbour at all. The federal government sets no food maximum level for most of them either, leaving the operative dietary references to European regulators. This briefing is precise about the space between naming a hazard and limiting it, because that space is where a certification does its job or fails to.

California is the one American jurisdiction that has looked at these metals and written down, in law, that they cause cancer or reproductive harm. Under Proposition 65 the state lists lead, cadmium, inorganic arsenic, hexavalent chromium, and nickel as chemicals known to it to cause cancer, reproductive toxicity, or both [1]. That is a formal governmental judgment about hazard, made by the state's own qualified experts, and it is the opposite of a data gap. The gap is what comes after the judgment. Having named the hazard, California sets no limit for any of these metals in food. The purpose of this briefing is to be precise about why, and about what the space between "named as a hazard" and "limited in food" actually contains, because that space is where a certification either does its job or fails to.

1 | What Proposition 65 is, and is not

Proposition 65, the Safe Drinking Water and Toxic Enforcement Act of 1986, is a right-to-know statute [2]. It does one thing: it requires the state to maintain a list of chemicals known to cause cancer or reproductive toxicity, and it requires a business to warn before it exposes a person to a listed chemical above a safe-harbour level. For a carcinogen that level is a No Significant Risk Level, set at the dose carrying a one-in-100,000 lifetime cancer risk; for a reproductive toxicant it is a Maximum Allowable Dose Level, set a thousandfold below the no-observed-effect level [3]. Both are warning triggers expressed as a daily intake in micrograms per day. Neither is, or was ever meant to be, a maximum permitted concentration of the metal in a food. The

statute governs disclosure, not composition. A product can carry any concentration of a listed metal and remain compliant, provided the exposure stays under the safe-harbour or the warning is given.

That distinction is not pedantry. It is the whole of the matter, because it means a Proposition 65 listing is a statement about a hazard that deliberately declines to answer the question a food buyer actually needs answered: how much is too much in the product itself.

2 | Where the state stops

Set the five listings against their safe-harbours and the pattern is stark [1][3]. For lead, California has adopted an oral Maximum Allowable Dose Level of 0.5 micrograms per day and a cancer No Significant Risk Level of 15 micrograms per day. For cadmium it has adopted an oral Maximum Allowable Dose Level of 4.1 micrograms per day, on the developmental endpoint. For hexavalent chromium it has adopted an oral Maximum Allowable Dose Level of 8.2 micrograms per day. These three at least carry a number a business can measure a diet against. For inorganic arsenic and for nickel, California has adopted no oral safe-harbour at all: nickel's only Proposition 65 numbers are two inhalation cancer levels for refinery dust and subsulfide, and there is no oral figure for elemental nickel or nickel compounds as a class [4]. The metal is on the list for developmental and reproductive harm, added in 2018 after the state's own hazard-identification document found reduced offspring viability and body weight the most consistent effects of prenatal nickel exposure [5]; and then the trail ends. Nothing translates that finding into a dietary threshold.

So even at its most complete, Proposition 65 gives a business a labelling trigger, not a food limit, and at its least complete it gives nothing an eater can use at all. The state that was willing to name the hazard was not, by the design of the instrument it used, ever going to set the limit.

3 | The federal silence behind it

California's reticence would matter less if the federal government had filled the space, but it has not. The United States sets no maximum level for nickel, for cadmium, for tin, or for total or hexavalent chromium in food; its enforceable numbers reach only lead and inorganic arsenic, and only in a short list of infant categories through the Closer to Zero action levels. For the metals Proposition 65 names, the operative dietary reference values are not American at all. They are the European Food Safety Authority's tolerable intakes, and the binding maximum levels are the European Union's, set for lead, cadmium, inorganic arsenic, and tin under Regulation (EU) 2023/915 and for nickel under Regulation (EU) 2024/1987 [6][7]. EFSA set the tolerable daily intake for nickel at 13 micrograms per kilogram of body weight and then reported that ordinary European diets already exceed it, toddlers and young children most of all [8]; it set the tolerable weekly intake for cadmium at 2.5 micrograms per kilogram and found average diets sitting against it [9]. Those are the numbers that actually answer the composition question, and none of them is Californian, or American.

The result is a jurisdiction that named five metals as hazards and left the concentration question to a European regulator, and a federal system that mostly did not ask it. For a brand selling into California and shipping into the EU, the exposure is not hypothetical: the certificate that says nothing about nickel is silent on a metal California calls a reproductive toxicant and the EU caps by law.

4 | The gap is the mandate

This is the space the Heavy Metal Tested and Certified program was built to occupy, and it is worth stating without embellishment. Where a government has judged a metal a hazard but set no dietary limit, HMTc does

not invent one. It adopts the strictest maximum level any credible government has set for that product and analyte anywhere in the world, converts it to the product's native basis, and certifies each lot against it. Where a government limit exists, the standard cites it; where the only sovereign act is a hazard listing with no number attached, the program supplies the number the listing implies but never states, drawn from the tightest limit a peer regulator has been willing to set. The listing says the hazard is real. The certification says how much is too much, and holds the product to it.

A Proposition 65 listing, in the end, is a government telling the public that a metal is dangerous and then handing the public a warning label instead of a limit. It is an honest half of the job. The other half — the number, measured the same way every time, published in advance, traceable to the sovereign act behind it — is the half a certification exists to do. The metals are already named. The gap is already there. What remains is to close it.

References

[1] Proposition 65 List of Chemicals Known to the State to Cause Cancer or Reproductive Toxicity
California Office of Environmental Health Hazard Assessment (OEHHA), current list; lead, cadmium, inorganic arsenic compounds, hexavalent chromium, and nickel/nickel compounds all listed.
GOVERNMENT

[2] Safe Drinking Water and Toxic Enforcement Act of 1986 (Proposition 65)
California Health & Safety Code §25249.5 et seq. A right-to-know warning statute; establishes a listing and warning obligation, not a food composition standard.
STATUTE

[3] Proposition 65 Safe Harbor Levels — No Significant Risk Levels and Maximum Allowable Dose Levels
OEHHA safe-harbour tables: lead oral MADL 0.5 µg/day and cancer NSRL 15 µg/day; cadmium oral MADL 4.1 µg/day; hexavalent chromium oral MADL 8.2 µg/day. Warning triggers in µg/day, not food maximum levels.
GOVERNMENT

[4] Nickel and Nickel Compounds — Proposition 65 Warnings Website fact sheet
OEHHA, February 2025. Listing scope and safe-harbour status; the only nickel safe-harbours are inhalation cancer NSRLs (Nickel Refinery Dust 0.8 µg/day, Nickel Subsulfide 0.4 µg/day). No oral safe-harbour for elemental nickel or nickel compounds.
GOVERNMENT

[5] Evidence on the Developmental and Reproductive Toxicity of Nickel and Nickel Compounds
OEHHA hazard-identification document, July 2018, underlying the soluble-nickel-compounds reproductive-toxicity listing effective 26 October 2018. Critical findings: reduced offspring viability and body weight from prenatal exposure.
GOVERNMENT

[6] Commission Regulation (EU) 2023/915 on maximum levels for certain contaminants in food
European Commission, 2023 (consolidated). Binding food maximum levels for lead, cadmium, inorganic arsenic, and tin, including infant and child food categories.
REGULATION

[7] Commission Regulation (EU) 2024/1987 amending (EU) 2023/915 as regards maximum levels of nickel in food
European Commission, 2024. The EU's first binding nickel maximum levels in food; most entries apply from 1 July 2025, cereal limits from 1 July 2026.
REGULATION

[8] Update of the Risk Assessment of Nickel in Food and Drinking Water
EFSA Panel on Contaminants in the Food Chain (CONTAM), EFSA Journal 2020;18(11):6268. Chronic TDI 13 µg Ni/kg body weight/day; mean dietary exposure exceeds it, children most.
GOVERNMENT

[9] Cadmium in food — Scientific Opinion of the Panel on Contaminants in the Food Chain
EFSA CONTAM Panel, EFSA Journal 2009;7(10):980. Tolerable weekly intake of 2.5 µg Cd/kg body weight; average dietary exposure at or near the TWI.
GOVERNMENT

STATUS AND LICENCE

Editorially reviewed in-house; no external peer review has been completed on this briefing. Published as a preprint with its review state tracked on the article page. Released under CC BY 4.0.

COMPETING INTERESTS

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DATA AVAILABILITY

The works this briefing draws on are listed in full in the references above and on the briefing page at https://-heavymetalcertified.com/briefings/the-prop-65-gap. Primary legal instruments and expert-body assessments are named with their public identifiers; the underlying occurrence and toxicology corpus is maintained on the Heavy Metal Index.

PUBLICATION HISTORY

This revision 2026-08. The full dated record is maintained at https://heavymetalcertified.com/briefings/the-prop-65-gap.